PROJECT PROGRESS REPORT

PROJECT PROGRESS REPORT
(Reporting No. 07/ 10)
Reporting Period: From 27/08/2026 To 03/09/2026

1. Student Information

Student Name	MAN VISHNUBHAI PATEL
Enrollment Number	25IC03CA053
Program/Branch	MCA / ICA
Semester	3
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2. Project Details

Type	Research Project
Department/Domain	MCA (DS/ML) /AI &ML
Mentor Name	Mr. Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Project Title

- OpenCV - Based System for Crack Identification and Width Estimation

4. Progress During the Reporting Period

A. Previous Reporting Review

Planned Task	Status	Remarks
Install & Import Libraries	**Completed**	Required libraries were installed and imported successfully for image processing and crack analysis.
Project Configuration	**Completed**	Project configuration, output directories, and OpenCV parameters were successfully configured.
Output & Mask Directories	**Completed**	Directories for processed images, crack masks, and annotated images were created successfully.
OpenCV Parameters	**Completed**	Gaussian blur, black-hat transformation, morphological processing, and crack-area parameters were configured and tested.
Pixel-to-mm Calibration	**Completed**	Pixel-to-mm calibration was configured and integrated with crack-width estimation.
Image Upload	**Completed**	Image upload functionality was implemented and tested for crack-image analysis.
Image Preprocessing	**Completed**	Image preprocessing was implemented to improve image quality and prepare images for analysis.
Crack Segmentation	**Completed**	Crack regions were successfully segmented using image-processing and morphological techniques.
Crack Detection	**Completed**	Crack detection was implemented and tested on input images.
Crack Width Estimation	**Completed**	Crack-width estimation was implemented using detected crack regions and calibration settings.
Risk Classification	**Completed**	Risk classification was implemented based on the measured crack characteristics and defined thresholds.
Result Visualization	**Completed**	Annotated results were generated to display detected cracks and measurement information.
CSV Report	**Pending**	CSV report generation is remaining and will be implemented in the next development phase.
ZIP Download	**Pending**	ZIP download functionality for project outputs is remaining and will be implemented after CSV report generation.

B. Work Completed During This Reporting Period

- **Activities/Tasks Completed:** Completed risk classification and result visualization, integrated the developed modules, and performed testing on multiple crack images.
- **Modules/Features Developed:** Developed and integrated risk classification and annotated result visualization features with the existing crack detection and width estimation modules.
- **Research/Literature Work Completed:** Reviewed research papers related to crack severity classification, crack-width measurement, image processing, and computer-vision-based structural crack analysis.
- **Design/Development/Testing Activities:** Improved the image-analysis workflow, tested crack detection and width estimation results, checked calibration settings, and resolved implementation issues.
- **Results/Outputs Achieved:** Successfully generated crack detection, crack-width measurement, risk classification, and annotated image results; CSV report generation and ZIP download remain pending.

C. Evidence / Deliverables Produced

Sr. No.	Deliverable / Evidence	Description / Link / Attachment
1	Risk Classification Output	Generated crack risk classification results based on the measured crack characteristics.
2	Result Visualization	Annotated images showing detected crack regions, measurements, and risk status.
3	Crack Detection Results	Tested outputs showing successfully identified crack regions in input images.
4	Crack Width Estimation Results	Generated crack-width measurements using the implemented calibration settings.
5	Testing Evidence	Screenshots and outputs obtained from testing the integrated crack-analysis modules.
6	Updated Source Code	Updated Google Colab code containing the completed crack analysis, measurement, risk classification, and visualization modules.
7	CSV Report	CSV generation will be completed.
8	ZIP Download	**Pending** — ZIP download of generated results will be completed after CSV report implementation.

Original Image:



Quick result table generated

index	Image_Name	Crack_Detected	Crack_Regions	Maximum_Width_MM	Average_Width_MM	Crack_Length_MM	Severity	Risk_Level	Health_Status
0	201.jpg	YES	50	2.718	0.702	1282.627	HEAVY	HIGH	HEAVY

Width Formula Verification

```
Distance Transform Value = 3.5 pixels
Width = 2 x 3.5
Width = 7.0 pixels
Width = 7.0 / 10.0
Actual Width = 0.700 mm

 Width formula verification completed.
```

Crack Measurement Points

	Image_Name	Crack_ID	Point_ID	X	Y	Distance_Pixels	Width_Pixels	Width_MM	Severity	Risk_Level
0	201.jpg	1	1	113	0	2.0000	4.0000	0.4000	HEAVY	HIGH
1	201.jpg	1	2	129	0	2.0000	4.0000	0.4000	HEAVY	HIGH
2	201.jpg	1	3	188	0	2.0000	4.0000	0.4000	HEAVY	HIGH
3	201.jpg	1	4	252	0	2.0000	4.0000	0.4000	HEAVY	HIGH
4	201.jpg	1	5	311	0	2.0000	4.0000	0.4000	HEAVY	HIGH
5	201.jpg	1	6	345	0	2.0000	4.0000	0.4000	HEAVY	HIGH
6	201.jpg	1	7	367	0	2.0000	4.0000	0.4000	HEAVY	HIGH
7	201.jpg	1	8	31	1	3.0000	6.0000	0.6000	HEAVY	HIGH
8	201.jpg	1	9	32	1	3.0000	6.0000	0.6000	HEAVY	HIGH
9	201.jpg	1	10	33	1	3.0000	6.0000	0.6000	HEAVY	HIGH
10	201.jpg	1	11	34	1	3.0000	6.0000	0.6000	HEAVY	HIGH
11	201.jpg	1	12	35	1	3.0000	6.0000	0.6000	HEAVY	HIGH
12	201.jpg	1	13	36	1	3.0000	6.0000	0.6000	HEAVY	HIGH
13	201.jpg	1	14	37	1	3.0000	6.0000	0.6000	HEAVY	HIGH
14	201.jpg	1	15	38	1	3.1969	6.3938	0.6394	HEAVY	HIGH
15	201.jpg	1	16	39	1	3.5969	7.1938	0.7194	HEAVY	HIGH
16	201.jpg	1	17	111	1	3.5969	7.1938	0.7194	HEAVY	HIGH
17	201.jpg	1	18	112	1	2.8000	5.6000	0.5600	HEAVY	HIGH
18	201.jpg	1	19	128	1	2.1969	4.3938	0.4394	HEAVY	HIGH
19	201.jpg	1	20	188	1	2.0000	4.0000	0.4000	HEAVY	HIGH

D. Current Project Progress

Overall Project Completion: 95%

Sr. No.	Project Component	Status	Completion
1	Library Installation	Completed	100%
2	Import Libraries	Completed	100%
3	Project Configuration	Completed	100%
4	Image Upload	Completed	100%
5	Image Preprocessing	Completed	100%
6	Crack Segmentation	Completed	100%
7	Crack Detection	Completed	100%
8	Crack Width Estimation	Completed	100%
9	Risk Classification	Completed	100%
10	Result Visualization	Completed	100%
11	CSV Report	Completed	100%
12	ZIP Download	**Working**	**20%**

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E. Challenges / Issues Faced

- Variations in crack images, such as different lighting conditions, shadows, surface textures, image quality, and crack patterns, affected the consistency of image analysis.
- Selecting suitable OpenCV parameters for Gaussian filtering, black-hat transformation, morphological operations, and crack-area filtering required repeated testing and adjustment.
- Some thin, broken, or low-contrast cracks were difficult to distinguish from background noise and concrete surface patterns.
- Accurate crack-width estimation and pixel-to-mm calibration required proper calibration settings and testing with different crack images.
- During system integration, maintaining consistent results between crack detection, measurement, risk classification, and visualization modules required additional debugging and testing.

- Testing with multiple images revealed variations in detection and measurement results, requiring parameter optimization and result validation.
- The final ZIP Download functionality is still pending and needs to be integrated with the generated annotated images, crack masks, processed images, and CSV report.

F. Actions Taken / Solution Implemented

- Improved crack-image preprocessing using grayscale conversion, Gaussian filtering, black-hat transformation, thresholding, and morphological operations to reduce noise and enhance crack regions.
- Tuned OpenCV parameters such as Gaussian kernel, black-hat kernel, morphological kernel, minimum crack area, and closing iterations to improve detection consistency.
- Implemented crack segmentation and detection to identify crack regions from processed images.
- Implemented skeletonization-based crack analysis for extracting crack structure and supporting length and width measurement.
- Integrated pixel-to-mm calibration with crack-width estimation to obtain measurable crack dimensions.
- Implemented and refined risk classification using defined crack-width/severity thresholds.
- Developed annotated result visualization to display detected crack regions and measurement information clearly.
- Tested the complete analysis workflow with multiple crack images, reviewed outputs, and corrected processing issues found during testing.
- Completed the CSV report generation for storing image-wise crack analysis results.
- ZIP Download functionality remains the only pending feature and will be completed in the next reporting period.

G. Plan for Next Reporting

Sr. No.	Planned Activity	Expected Deliverable	Target Date
1	Complete ZIP Download functionality	ZIP file containing annotated images, crack masks, processed images, and CSV report	17 September 2026
2	Perform final system testing and error handling	Fully tested and stable integrated system	17 September 2026
3	Complete final validation, documentation, and project demonstration	Final validated system, complete project report, and demonstration-ready project	24 September 2026

H. Learning / Skills Acquired

- **Technical Skills:** Improved practical skills in crack segmentation, detection, width estimation, pixel-to-mm calibration, risk classification, and result visualization.
- **Programming Tools:** Strengthened skills in Python, OpenCV, NumPy, Pandas, Matplotlib.
- **Frameworks:** Gained practical experience using OpenCV and Scikit-image for image processing and skeletonization.
- **Research Skills:** Improved literature review and analysis skills related to computer-vision-based crack detection and structural crack measurement.
- **Problem-Solving Skills:** Learned to identify and resolve issues related to image noise, thin cracks, parameter selection, segmentation, and measurement accuracy.
- **Testing Skills:** Gained experience in testing the integrated system with multiple crack images and analysing the generated outputs.
- **Documentation Skills:** Improved technical documentation, result recording, progress reporting, and project report preparation.
- **Teamwork / Communication Skills:** Improved communication and presentation skills through project discussions, progress reviews, and explanation of technical development.

4. Tasks Accomplished

- Completed the Risk Classification module based on crack-width measurements and defined severity thresholds.
- Completed Result Visualization with annotated crack images and measurement information.
- Completed CSV Report generation for storing crack analysis results and measurements.
- Integrated the completed modules into the overall crack identification and width estimation workflow.
- Performed testing and validation using multiple crack images and reviewed the generated outputs.
- Improved system performance through parameter tuning, debugging, and error handling.
- Verified the pixel-to-mm calibration and crack-width estimation results.
- Completed the major project components; only ZIP Download functionality remains pending.

5. Tasks in Progress / Next Plan

- Complete the ZIP Download functionality to package all generated project outputs into a single downloadable ZIP file.
- Perform final system testing and error handling after integrating the ZIP download feature.
- Validate the complete workflow from image upload to crack detection, width estimation, risk classification, visualization, CSV report, and ZIP download.
- Finalize project optimization and documentation based on the testing results.
- Prepare the final project demonstration, presentation, and viva.

6. Learnings / Skills Acquired

- Learned to design and organize a complete project workflow from individual modules to final output.
- Improved code integration and modular development skills by connecting multiple project components.
- Gained experience in interpreting project results and identifying areas requiring improvement.

- Learned effective debugging and error-resolution approaches during development and testing.
- Improved understanding of project planning, progress tracking, and milestone management.
- Developed skills in preparing and presenting technical project information clearly and systematically.

References:

- **Hassouna, M., Marzouk, M., & Fathalla, E. (2026).** *Automated Low-Cost Framework for Crack Measurements in RC Structures Using Deep Learning Approach.* **Scientific Reports, 16, 14678.**
- **Chen, W., Peng, Z., Chen, X., Zhao, L., Xu, T., Li, Q., Huang, X., & Kannangara, K. K. P. M. (2026).** *Concrete Crack Detection and Classification Methods Based on Machine Vision and Deep Learning.* **Sensors, 26(8), 2381.**
- **Bodke, K., Bhirud, S., & Sangle, K. K. (2026).** *Enhanced Crack Width and Depth Measurement through Binary Image Processing and Geometric Analysis.* **Scientific Visualization, 18(1), 49–61.**

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks
Progress as per Previous Plan	L3– Apply	All planned tasks completed	Most completed	Partial completion	Very little progress	
Technical / Research Implementation	L4– Analyze	Excellent implementation with clear evidence	Good implementation	Basic implementation	Insufficient evidence	
Quality of Deliverables	L5– Evaluate	High-quality, validated deliverables	Good quality	Acceptable	Poor / incomplete	
Problem Solving & Challenges	L4– Analyze	Challenges clearly identified and effectively solved	Most issues addressed	Limited solutions	No clear effort	

Planning for Next Phase	L6–Create	Clear, realistic and measurable plan	Mostly clear plan	Basic plan	No proper plan	
Documentation & Presentation	L3–Apply	Well-structured and professional	Minor issues	Average clarity	Poor documentation	
Total marks out of 100						

Evaluation

Student Signature	Mentor (Institute) Signature

Mentor (Institute) Remarks

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